

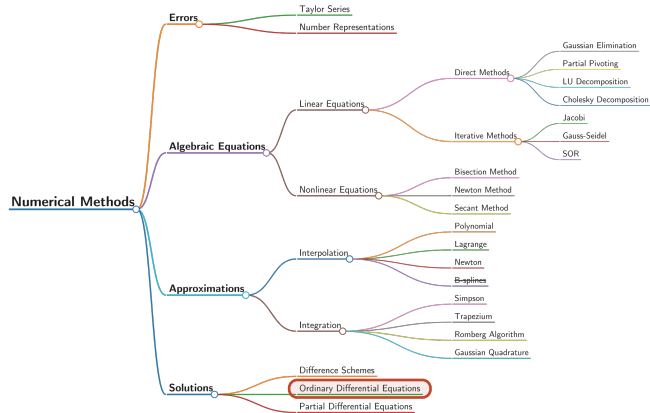
Numerical Methods for ODEs

CTMS-MAT-13 • Forward Euler, Backward Euler, Crank–Nicolson, Heun

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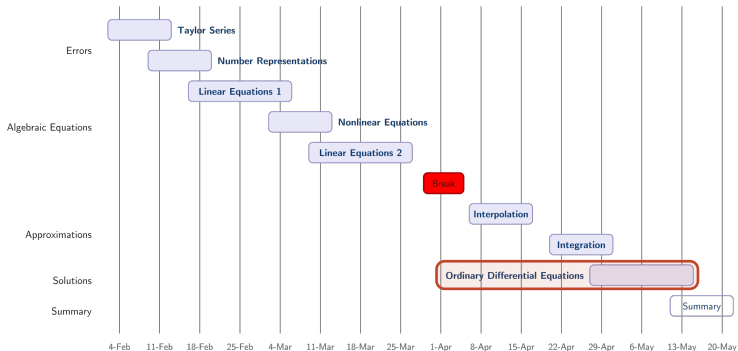
Where this sits in the course



Full outline: djps.github.io/docs/numericalmethods/intro/intro/

Schedule

Schedule 2026



Notes: djps.github.io/teaching/numerical_methods.pdf

Today

The problem

Ordinary differential equations

An **ordinary differential equation** relates an unknown function, its parameter, and its derivative with respect to that parameter:

$$\frac{dx}{dt} = x^2 + \sin(x(t)), \quad t \in [0, T], \quad x(t) \in \mathbb{R}.$$

Here $\dot{x}(t) = f(x(t)) = x^2 + \sin x$: unknown x , parameter t , derivative $\dot{x}$.

Notes: [section 9](#) • [website](#)

Two problem types, two method families

What is given

IVP an initial condition, e.g.
 $y' = -\sin y$ with $y(0)$
given;

BVP boundary values, e.g. $y(0)$
and $y(L)$ given.

What the scheme uses

one-step: $x_n \longrightarrow x_{n+1}$

multi-step: $x_n, x_{n-1} \longrightarrow x_{n+1}$

A scheme depending on the previous value *only* is a **one-step** method.

TAKEAWAY

Everything in this lecture is a one-step method for an IVP: march forward in t , one step of size h at a time.

The four schemes

Forward Euler

Solve $\dot{x}(t) = f(x, t)$, set $t_n = nh$ and $x(t_n) \approx u_n$.

From the *forward* difference,

$$x'(t) \approx \frac{x(t+h) - x(t)}{h} \quad \Longrightarrow \quad f_n = \frac{u_{n+1} - u_n}{h},$$

$$u_{n+1} = u_n + hf_n, \quad f_n = f(t_n, u_n).$$

Everything on the right hand side is known: the scheme is **explicit**.

Backward Euler

From the *backward* difference, $f_{n+1} = (u_{n+1} - u_n)/h$:

$$u_{n+1} = u_n + hf_{n+1}, \quad f_{n+1} = f(t_{n+1}, u_{n+1}).$$

The unknown appears on both sides — the scheme is **implicit** and each step costs work (e.g. Newton's method).

Linear case $\vec{x}' = A\vec{x}$

$$\vec{u}_{n+1} = \vec{u}_n + hA\vec{u}_{n+1} \implies (I - hA) \vec{u}_{n+1} = \vec{u}_n \implies \vec{u}_{n+1} = (I - hA)^{-1} \vec{u}_n.$$

One linear solve per step.

Crank–Nicolson and Heun

Crank–Nicolson —average the two Eulers:

$$u_{n+1} = u_n + \frac{h}{2}(f_n + f_{n+1})$$

- implicit, second order;
- the standard choice for heat equations.

Heun —predict, then average:

$$u_{n+1} = u_n + \frac{h}{2}\left(f_n + f(u_n + hf_n, t_{n+1})\right)$$

- the **predictor** $u_n + hf_n$ is one forward Euler step;
- explicit, second order.

TAKEAWAY

Heun buys Crank–Nicolson’s accuracy without the implicit solve —by estimating f_{n+1} rather than solving for it.

Explicit or implicit

Explicit — u_{n+1} computed directly.

- Forward Euler
- Heun

Cheap per step, restricted step size.

Implicit — u_{n+1} appears inside f .

- Backward Euler
- Crank–Nicolson

Costly per step, generous step size.

The rest of the lecture is about which trade wins, and when.

The same schemes are quadrature rules

Since $f = y'$,

$$y(t+h) = y(t) + \int_t^{t+h} f(y(s), s) ds,$$

and every scheme above is a rule for that integral:

- Forward Euler $\equiv$ left Riemann sum;
- Backward Euler $\equiv$ right Riemann sum;
- Crank–Nicolson $\equiv$ trapezoidal rule.

TAKEAWAY

Nothing new is being invented: the accuracy of an ODE scheme is the accuracy of its quadrature rule.

Accuracy

One-step methods and truncation error

A general one-step method reads

$$u_{n+1} = u_n + h \phi(t_n, u_n, f_n, h),$$

with ϕ the **increment function** —for forward Euler, $\phi = f_n$.

Substituting the exact solution leaves a residual $\varepsilon_n = h\tau_n$:

$$\tau_n = \frac{y_{n+1} - y_n}{h} - f(t_n, y_n), \quad \tau(h) = \max_n |\tau_n|.$$

τ_n is the **local** truncation error; $|y(t_n) - u_n|$ is the **global** one.

Order of a method

Taylor expansion gives, for forward Euler,

$$\frac{y_{n+1} - y_n}{h} = y'_n + \frac{h}{2}y''(\xi) \implies \tau(h) \leq \frac{h}{2} \max_{t \in (t_0, T)} |y''(t)| = \mathcal{O}(h).$$

Order

A one-step method has **order** p if $\tau(h) = \mathcal{O}(h^p)$ as $h \rightarrow 0$.

Forward and backward Euler are order 1; Heun and Crank–Nicolson are order 2.

Stability

Zero-stability

Definition

The method $u_{n+1} = u_n + h\phi(t_n, u_n, f_n, h)$ is **zero-stable** if there is an $h_0 > 0$ such that for all $h \in [0, h_0]$ and all perturbations with $|\delta_n| \leq \varepsilon$, the perturbed sequence $z_{n+1} = z_n + h\phi(t_n, z_n, f(z_n, t_n), h) + \delta_{n+1}$, $z_0 = y_0 + \delta_0$, satisfies $|z_n - u_n| \leq C\varepsilon$.

In words: small perturbations during the computation produce only small perturbations in the answer —the fixed- h , finite- T notion of stability.

Absolute stability

Now fix $h > 0$ and ask what happens as $t_n \rightarrow \infty$. Use the linear test problem

$$y'(t) = \lambda y(t), \quad y(0) = 1, \quad \lambda \in \mathbb{C}, \quad y(t) = e^{\lambda t},$$

which decays exactly when $\operatorname{Re}(\lambda) < 0$.

Definition

A scheme is **absolutely stable** if $|u_n| \rightarrow 0$ as $t_n \rightarrow \infty$. Its **region of absolute stability** is

$$A = \{h\lambda \in \mathbb{C} : |u_n| \rightarrow 0 \text{ as } t_n \rightarrow \infty\}.$$

The four regions

Forward Euler

$$u_n = (1 + h\lambda)^n u_0$$

stable iff $|1 + h\lambda| < 1$: the unit disc centred at -1 .

Heun

$$u_n = \left(1 + h\lambda + \frac{(h\lambda)^2}{2}\right)^n u_0$$

larger than forward Euler's, still bounded.

Backward Euler

$$u_n = \frac{u_0}{(1 - h\lambda)^n}$$

stable iff $|1 - h\lambda| > 1$: everything *outside* the disc at $+1$.

Crank–Nicolson

$$u_n = \left(\frac{1 + \frac{1}{2}h\lambda}{1 - \frac{1}{2}h\lambda}\right)^n u_0$$

stable for all h whenever $\text{Re}(h\lambda) < 0$.

TAKEAWAY

Runge–Kutta methods

Runge–Kutta methods

One step, but several **stages**:

$$u_{n+1} = u_n + h F(t_n, u_n, h, f), \quad F = \sum_{i=1}^s b_i K_i,$$
$$K_i = f \left(u_n + h \sum_{j=1}^s a_{ij} K_j, t_n + c_i h \right).$$

The scheme is fixed by $\vec{b}, \vec{c} \in \mathbb{R}^s$ and $A \in \mathbb{R}^{s \times s}$, collected in a **Butcher array**:

$$\begin{array}{c|c} \vec{c} & A \\ \hline & \vec{b}^\top \end{array}$$

If $a_{ij} = 0$ for $j \geq i$ the stages unwind one after another and the method is explicit.

0	0	0	0	0
1/2	1/2	0	0	0
1/2	0	1/2	0	0
1	0	0	1	0
	1/6	1/3	1/3	1/6

$$c_i = \sum_j a_{ij} \text{ and } \sum_i b_i = 1.$$

$$K_1 = f(u_n, t_n)$$

$$K_2 = f(u_n + \frac{h}{2} K_1, t_n + \frac{h}{2})$$

$$K_3 = f(u_n + \frac{h}{2} K_2, t_n + \frac{h}{2})$$

$$K_4 = f(u_n + h K_3, t_n + h)$$

$$u_{n+1} = u_n + h \left(\frac{K_1}{6} + \frac{K_2}{3} + \frac{K_3}{3} + \frac{K_4}{6} \right)$$

Consistency and the limits of order

The local truncation error is $h\tau_{n+1}(h) = y_{n+1} - y_n - hF(t_n, u_n, h, f)$, and $\tau(h) \rightarrow 0$ if and only if $\sum_i b_i = 1$.

- An s -stage *explicit* scheme has order at most s ;
- an *implicit* scheme reaches order $2s$;
- no s -stage explicit scheme attains order s for $s \geq 5$.

Schemes are derived by matching the Taylor expansion of u_{n+1} at t_n with that of $y(t_{n+1})$ —as follows.

Deriving a 2-stage explicit scheme

With $s = 2$ and A strictly lower triangular,

$$K_1 = f(y_n, t_n), \quad K_2 = f(y_n + h a_{21} K_1, t_n + h c_2),$$

$$K_2 = f_n + h c_2 \left(\frac{\partial f}{\partial y} f + \frac{\partial f}{\partial t} \right) + \mathcal{O}(h^2),$$

so

$$u_{n+1} = u_n + h(b_1 + b_2)f_n + h^2 b_2 c_2 \left(\frac{\partial f}{\partial y} f + \frac{\partial f}{\partial t} \right) + \mathcal{O}(h^3).$$

Matching against $y_{n+1} = y_n + h f_n + \frac{h^2}{2}(f_y f + f_t) + \mathcal{O}(h^3)$:

$$b_1 + b_2 = 1, \quad b_2 c_2 = \frac{1}{2}.$$

Heun, recovered

Choose $b_1 = b_2 = \frac{1}{2}$ and $c_2 = a_{21} = 1$:

$$\begin{array}{c|cc} 0 & & \\ 1 & 1 & \\ \hline & \frac{1}{2} & \frac{1}{2} \end{array} \quad \Longrightarrow \quad u_{n+1} = u_n + \frac{h}{2} (f_n + f(u_n + hf_n, t_n + h)),$$

which is exactly **Heun's method**.

TAKEAWAY

The order conditions have many solutions —every classical explicit scheme is one particular choice of Butcher array.

Order tells you how fast,
stability tells you whether.

Summary

- An IVP is marched forward in steps of h ; a one-step method uses only u_n .
- Forward Euler and Heun are explicit; backward Euler and Crank–Nicolson are implicit and cost a solve per step.
- Every scheme is a quadrature rule for $\int_t^{t+h} f \, ds$.
- Order p means $\tau(h) = \mathcal{O}(h^p)$: Euler 1, Heun and CN 2, RK4 4.
- Absolute stability bounds h for explicit schemes; implicit schemes cover the left half-plane.

Take-home messages

1. **Choose the scheme for the problem, not the accuracy table.** A stiff problem makes an explicit method's step size, not its order, the binding constraint.
2. **Implicit costs are per step; explicit costs are per unit time.** Compare total work to a fixed tolerance, not work per step.
3. **Order is asymptotic.** $\mathcal{O}(h^4)$ says nothing until h is small enough for the expansion to be valid.
4. **Runge–Kutta is a recipe, not a scheme.** Give the Butcher array and the method is completely specified.